

Covariance-Method Claims Are Ensemble-Relative: Named Measures and an Audit Program

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Abstract

A claim about the behavior of a covariance or correlation method is a claim about a distribution over matrices, and the distribution is rarely stated. Papers validate estimators, factorizations, and portfolio constructions on whatever generator is at hand: a Wishart draw, a uniform LKJ sample, one empirical panel. We measure how much this omission can matter. Across fifteen named ensembles at identical dimension and sample size, the Ledoit–Wolf shrinkage intensity ranges from 0.08 to 0.87; a probability algorithm whose median error is 10^{-5} under twelve ensembles degrades by two orders of magnitude under kernel-generated fields, for a stated structural reason. The `randomcov` library names the measures and seeds the draws. The program this enables is an audit: re-run published covariance-method claims across the full set of named ensembles and report which verdicts survive and which flip. Two demonstrations are given here; the audits are the invitation.

1 The unstated measure

Empirical claims about covariance methods have the form “estimator A beats estimator B ,” “the shrinkage intensity is typically moderate,” “the factorization recovers structure,” with performance averaged over random test matrices. The average is with respect to some generating law, and the law is usually inherited from convenience: `numpy` Wishart draws, a uniform LKJ prior, a single equity panel. Different laws produce matrices with different spectra, different locality, and different factor content, so a verdict certified under one law carries no warrant under another. The omission is not a technicality, as the two measurements below show.

2 Fifteen named ensembles

`randomcov` generates correlation matrices under named, seeded measures: `lkj`, `onion`, `vine` (elliptope samplers); `wishart`, `spectrum` (spectral laws, including Marchenko–Pastur and spiked); `factor`, `hierarchical`, `block.equicorr` (low-rank and clustered structure); `ar1`, `kernel` (banded and locality-driven fields); `sparse.precision` (graphical-model structure); `archakov.hansen` (log-matrix parameterization); `residuals`, `walk`, `animals` (empirical-flavored and trajectory ensembles). Variance vectors are drawn separately, so correlation structure and scale can be crossed. Every draw is reproducible from a seed, which is what an audit needs.

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3 Demonstration: one estimator, twelve-fold spread

The Ledoit–Wolf shrinkage intensity is a workhorse diagnostic: how far the sample covariance is pulled toward its target. Fix $n = 30$ variables, $T = 60$ observations, eight replications, and vary only the generating ensemble.

ensemble	intensity	sd
animals	0.075	0.013
residuals	0.095	0.011
factor	0.104	0.013
archakov_hansen	0.117	0.015
kernel	0.140	0.091
walk	0.195	0.034
block_equicorr	0.261	0.097
vine	0.344	0.024
lkj	0.352	0.026
hierarchical	0.357	0.074
onion	0.363	0.035
ar1	0.364	0.250
spectrum	0.471	0.066
wishart	0.528	0.032
sparse_precision	0.869	0.067

The intensity spans 0.075 to 0.869 at identical (n, T) : a paper reporting “typical” shrinkage from one generator is reporting a property of its generator. The ordering is interpretable—strong low-rank structure (**animals**, **residuals**, **factor**) wants little shrinkage, near-unstructured draws (**wishart**, **sparse_precision**) want a great deal—and the within-ensemble standard deviation is itself informative: **ar1** spans 0.25 across its own parameter draws, so even a named ensemble can hide a range.

4 Demonstration: one algorithm, one failing ensemble

A deterministic algorithm for multinomial probit choice probabilities (the **winning** package; paper in that repository) reports median absolute error by ensemble after fitting a structured covariance grammar. Under twelve of the fifteen measures the error is $1\text{--}4 \times 10^{-5}$; under dense-strong ensembles it is $1\text{--}4 \times 10^{-4}$; under **kernel** fields (Matérn and RBF) it is 4×10^{-3} . The failure has a mechanism: kernel covariance is locality, approximately Markov, and offers a low-rank factorization no good purchase. A validation run on any single ensemble would have reported either “essentially exact” or “inadequate,” both correctly, for the same method.

5 The audit program

The protocol is short. Take a published claim about a covariance or correlation method; identify the generating law under which it was demonstrated; re-run the demonstration, seeded, under every named ensemble; report the verdict per ensemble; flag the flips and, where possible, the mechanism. Candidate claims, in rough order of exposure: shrinkage-target and intensity recommendations; hierarchical-risk-parity versus minimum-variance horse races; sparse precision

recovery guarantees; condition-number and regularization benchmarks; correlation-forecast comparisons. The two demonstrations above suggest what to expect: structure-sensitive methods will flip on structure-heavy ensembles, and locality is the ensemble property most likely to defeat spectral intuition.

This is a working paper and a program, not a completed audit: the demonstrations are ours, the audits are the invitation, and the library exists so that the next paper's claim can be tested across all fifteen measures in the time it currently takes to test one.